Grade 6 Accelerated
Unit 11 Assessment – Part 2
Calculator

Name____________________
Date_________________
Period_________

Showing Work with a Calculator: Because you are using a calculator, it is possible to accidently punch the wrong button. To ensure you earn partial credit, show the operation that you entered into the calculator and the result. It is helpful if you label the operation with information from the word problem.

1. Florida wildlife researchers used trail cameras in two different nature parks in South Florida to estimate the population of white-tailed deer in the Florida Panther National Wildlife Refuge and the Bear Island Unit of the Big Cypress National Preserve. The table shows the average number of deer per square mile in each area and the total area of each park.

Nature Park	**Average Number of White-tailed Deer per Square Mile**	**Total Area of Park (square miles)**
Florida Panther National Wildlife Refuge	16.64	$41\frac{1}{3}$
Big Cypress National Preserve	13.75	$66\frac{11}{20}$

Determine the estimated total population of deer in the two nature parks. Show your steps for partial credit.

Total Deer Population	

2. The Cactus-to-Cloud Trail begins in Palm Springs, California, and ascends to San Jacinto Peak. The temperature, in degrees Celsius $^\circ\mathrm{C}$, in Palm Springs when Belinda began hiking the Cactus-to-Cloud Trail was $17\frac{7}{9}{}^\circ\mathrm{C}$. When she reached San Jacinto Peak, the temperature was $-4.4^\circ\mathrm{C}$.

What was the change in the temperature, in degrees Celsius, from Palm Springs to San Jacinto Peak? Show your steps for partial credit. Round to the nearest thousandth or write your answer as a fraction.

Change in Temperature	$^\circ\mathrm{C}$

3. The price list for K & J Homegrown Produce is shown.

Item	**Price**
Florida Green Cabbage	$\$0.49$ per pound
Yellow Squash	$\$1.29$ per pound
Florida Strawberries	$\$1.99$ per pound
Jalapeno Peppers	$\$0.99$ per pound

Tony purchased a cabbage that weighed $1\frac{2}{5}$ pounds. She purchased 2 pounds of squash, 5 pounds of strawberries and $\frac{1}{3}$ of a pound of peppers.

Determine the total amount, in dollars, of Tony's purchase. Show your steps for partial credit.

Total Amount	$

4. The table shows the departure from the average (the difference between the actual stock price and the average stock price for a given year) for a share of stock in a shoe company on the first business day of each month during 2020. A negative value represents the amount the share was below the average, and a positive value represents the amount the share was above the average. Prices are stated to the closest ten-thousandth of a dollar as stocks are often sold in large quantities.

Date	Departure from Average 2020 Stock Price	Date	Departure from Average 2020 Stock Price
January 2, 2020	-5.9759	July 1, 2020	-10.1899
February 3, 2020	-8.8566	August 3, 2020	-9.2707
March 2, 2020	-15.0828	September 1, 2020	7.2985
April 1, 2020	-28.3231	October 1, 2020	18.9869
May 1, 2020	-22.1221	November 2, 2020	14.7771
June 1, 2020	-8.0748	December 1, 2020	27.7037

The average stock price in 2020 for the shoe company was 106.4556.

a. What was the difference between the stock price on April 1, 2020 and the stock price on December 1, 2020? Show your steps for partial credit.

Difference in Stock Price	

b. Trevor bought $1{,}000$ shares of stock on January 2, 2020.

How much did Trevor pay for the $1{,}000$ shares? Round your answer to the nearest tenth. Show your steps for partial credit.

Cost of 1,000 shares on January 2, 2020	

5. A Dutch company built a submarine that can hold 10 passengers and 1 pilot. The submarine has a safety feature that will automatically raise the submarine if it goes below the certified depth of -650 feet. To show this feature to a buyer the submarine dove to a depth of $-872\frac{5}{12}$ feet. The automatic feature then raised the submarine $50\frac{3}{5}$ feet per minute.

   How much time, in minutes, does it take for the submarine to reach the certified depth? Round your answer to the nearest tenth of a minute. Show your steps for partial credit.

Time	minutes

6. The average emissions of CO_2 (in grams) per kilometer traveled across all types of passenger vehicles in Denmark for 2001 and 2019 is shown.

Year	**Average emissions of CO_2 (in grams) per kilometer**
2001	$174\frac{5}{12}$
2019	$107\frac{3}{32}$

In Denmark, the average distance traveled by a passenger vehicle in one month is 1167.6 kilometers.

Determine the difference in the average emissions of CO_2, in grams, of one passenger vehicle in one month in 2001 and one month in 2019. Show your steps for partial credit.

Difference in the Average Emissions of CO_2	grams